

ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

FINAL BASELINE MODEL: COMBINED SENTIMENT AND RATING PREDICTION MODEL

1. Introduction

The Final Baseline Model is developed by integrating multiple machine learning approaches to analyze student feedback more effectively. This model not only predicts the **sentiment (Positive, Negative, Neutral)** but also estimates a **rating score**, providing a more comprehensive understanding of feedback.

This combined approach improves the overall usefulness of the system compared to individual models.

2. Model Overview

The model consists of two main components:

- **Sentiment Classification Model**
- **Rating Prediction Model**

Both models use **TF-IDF vectorization** for feature extraction but apply different algorithms for their respective tasks.

3. Feature Representation

The textual feedback is converted into numerical format using **TF-IDF (Term Frequency–Inverse Document Frequency)**.

Key benefits:

- Identifies important words contributing to sentiment
- Reduces noise from common words
- Provides structured input for machine learning models

4. Sentiment Classification Model

For sentiment prediction, the model uses **Logistic Regression**.

Functionality:

- Classifies feedback into Positive, Negative, or Neutral
- Uses TF-IDF features as input
- Outputs probability for each class

5. Rating Prediction Model

For predicting rating values, the model uses **Random Forest Regressor**.

Functionality:

- Predicts a numerical rating score
- Uses ensemble learning (multiple decision trees)
- Produces stable and robust predictions

6. Working Pipeline

The complete pipeline works as follows:

1. Input feedback text
2. Text is cleaned and processed
3. TF-IDF converts text into numerical features
4. Features are passed to:
 - Logistic Regression → sentiment prediction
 - Random Forest → rating prediction
5. Final output includes both sentiment and rating

7. Model Performance

- Sentiment model achieved approximately **75% accuracy**
- Rating model achieved a reasonable prediction error (MSE around ~1)

Observations:

- Sentiment predictions are strong for positive feedback
- Rating predictions provide useful numerical insights
- Combined output gives better interpretation of feedback

RESULT

```
▶ predict_feedback("The faculty is very helpful but exams are tough")  
... ('Positive', np.float64(3.49))
```